

class09__mini__project__candy

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reading the data:

```
read.csv("candy-data.csv")
```

	competitorname	chocolate	fruity	caramel	peanutyalmondy	nougat
1	100 Grand	1	0	1	0	0
2	3 Musketeers	1	0	0	0	1
3	One dime	0	0	0	0	0
4	One quarter	0	0	0	0	0
5	Air Heads	0	1	0	0	0
6	Almond Joy	1	0	0	1	0
7	Baby Ruth	1	0	1	1	1
8	Boston Baked Beans	0	0	0	1	0
9	Candy Corn	0	0	0	0	0
10	Caramel Apple Pops	0	1	1	0	0
11	Charleston Chew	1	0	0	0	1
12	Chewey Lemonhead Fruit Mix	0	1	0	0	0
13	Chiclets	0	1	0	0	0
14	Dots	0	1	0	0	0
15	Dum Dums	0	1	0	0	0
16	Fruit Chews	0	1	0	0	0
17	Fun Dip	0	1	0	0	0
18	Gobstopper	0	1	0	0	0
19	Haribo Gold Bears	0	1	0	0	0
20	Haribo Happy Cola	0	0	0	0	0
21	Haribo Sour Bears	0	1	0	0	0
22	Haribo Twin Snakes	0	1	0	0	0
23	Hershey's Kisses	1	0	0	0	0
24	Hershey's Krackel	1	0	0	0	0
25	Hershey's Milk Chocolate	1	0	0	0	0
26	Hershey's Special Dark	1	0	0	0	0

27	Jawbusters	0	1	0	0	0
28	Junior Mints	1	0	0	0	0
29	Kit Kat	1	0	0	0	0
30	Laffy Taffy	0	1	0	0	0
31	Lemonhead	0	1	0	0	0
32	Lifesavers big ring gummies	0	1	0	0	0
33	Peanut butter M&M's	1	0	0	1	0
34	M&M's	1	0	0	0	0
35	Mike & Ike	0	1	0	0	0
36	Milk Duds	1	0	1	0	0
37	Milky Way	1	0	1	0	1
38	Milky Way Midnight	1	0	1	0	1
39	Milky Way Simply Caramel	1	0	1	0	0
40	Mounds	1	0	0	0	0
41	Mr Good Bar	1	0	0	1	0
42	Nerds	0	1	0	0	0
43	Nestle Butterfinger	1	0	0	1	0
44	Nestle Crunch	1	0	0	0	0
45	Nik L Nip	0	1	0	0	0
46	Now & Later	0	1	0	0	0
47	Payday	0	0	0	1	1
48	Peanut M&Ms	1	0	0	1	0
49	Pixie Sticks	0	0	0	0	0
50	Pop Rocks	0	1	0	0	0
51	Red vines	0	1	0	0	0
52	Reese's Miniatures	1	0	0	1	0
53	Reese's Peanut Butter cup	1	0	0	1	0
54	Reese's pieces	1	0	0	1	0
55	Reese's stuffed with pieces	1	0	0	1	0
56	Ring pop	0	1	0	0	0
57	Rolo	1	0	1	0	0
58	Root Beer Barrels	0	0	0	0	0
59	Runts	0	1	0	0	0
60	Sixlets	1	0	0	0	0
61	Skittles original	0	1	0	0	0
62	Skittles wildberry	0	1	0	0	0
63	Nestle Smarties	1	0	0	0	0
64	Smarties candy	0	1	0	0	0
65	Snickers	1	0	1	1	1
66	Snickers Crisper	1	0	1	1	0
67	Sour Patch Kids	0	1	0	0	0
68	Sour Patch Tricksters	0	1	0	0	0
69	Starburst	0	1	0	0	0

70	Strawberry bon bons	0	1	0	0	0
71	Sugar Babies	0	0	1	0	0
72	Sugar Daddy	0	0	1	0	0
73	Super Bubble	0	1	0	0	0
74	Swedish Fish	0	1	0	0	0
75	Tootsie Pop	1	1	0	0	0
76	Tootsie Roll Juniors	1	0	0	0	0
77	Tootsie Roll Midgies	1	0	0	0	0
78	Tootsie Roll Snack Bars	1	0	0	0	0
79	Trolli Sour Bites	0	1	0	0	0
80	Twix	1	0	1	0	0
81	Twizzlers	0	1	0	0	0
82	Warheads	0	1	0	0	0
83	Welch's Fruit Snacks	0	1	0	0	0
84	Werther's Original Caramel	0	0	1	0	0
85	Whoppers	1	0	0	0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent	win	percent
1		1	0	1		0		0.732	0.860	66.97173		
2		0	0	1		0		0.604	0.511	67.60294		
3		0	0	0		0		0.011	0.116	32.26109		
4		0	0	0		0		0.011	0.511	46.11650		
5		0	0	0		0		0.906	0.511	52.34146		
6		0	0	1		0		0.465	0.767	50.34755		
7		0	0	1		0		0.604	0.767	56.91455		
8		0	0	0		1		0.313	0.511	23.41782		
9		0	0	0		1		0.906	0.325	38.01096		
10		0	0	0		0		0.604	0.325	34.51768		
11		0	0	1		0		0.604	0.511	38.97504		
12		0	0	0		1		0.732	0.511	36.01763		
13		0	0	0		1		0.046	0.325	24.52499		
14		0	0	0		1		0.732	0.511	42.27208		
15		0	1	0		0		0.732	0.034	39.46056		
16		0	0	0		1		0.127	0.034	43.08892		
17		0	1	0		0		0.732	0.325	39.18550		
18		0	1	0		1		0.906	0.453	46.78335		
19		0	0	0		1		0.465	0.465	57.11974		
20		0	0	0		1		0.465	0.465	34.15896		
21		0	0	0		1		0.465	0.465	51.41243		
22		0	0	0		1		0.465	0.465	42.17877		
23		0	0	0		1		0.127	0.093	55.37545		
24		1	0	1		0		0.430	0.918	62.28448		
25		0	0	1		0		0.430	0.918	56.49050		
26		0	0	1		0		0.430	0.918	59.23612		

27	0	1	0	1	0.093	0.511	28.12744
28	0	0	0	1	0.197	0.511	57.21925
29	1	0	1	0	0.313	0.511	76.76860
30	0	0	0	0	0.220	0.116	41.38956
31	0	1	0	0	0.046	0.104	39.14106
32	0	0	0	0	0.267	0.279	52.91139
33	0	0	0	1	0.825	0.651	71.46505
34	0	0	0	1	0.825	0.651	66.57458
35	0	0	0	1	0.872	0.325	46.41172
36	0	0	0	1	0.302	0.511	55.06407
37	0	0	1	0	0.604	0.651	73.09956
38	0	0	1	0	0.732	0.441	60.80070
39	0	0	1	0	0.965	0.860	64.35334
40	0	0	1	0	0.313	0.860	47.82975
41	0	0	1	0	0.313	0.918	54.52645
42	0	1	0	1	0.848	0.325	55.35405
43	0	0	1	0	0.604	0.767	70.73564
44	1	0	1	0	0.313	0.767	66.47068
45	0	0	0	1	0.197	0.976	22.44534
46	0	0	0	1	0.220	0.325	39.44680
47	0	0	1	0	0.465	0.767	46.29660
48	0	0	0	1	0.593	0.651	69.48379
49	0	0	0	1	0.093	0.023	37.72234
50	0	1	0	1	0.604	0.837	41.26551
51	0	0	0	1	0.581	0.116	37.34852
52	0	0	0	0	0.034	0.279	81.86626
53	0	0	0	0	0.720	0.651	84.18029
54	0	0	0	1	0.406	0.651	73.43499
55	0	0	0	0	0.988	0.651	72.88790
56	0	1	0	0	0.732	0.965	35.29076
57	0	0	0	1	0.860	0.860	65.71629
58	0	1	0	1	0.732	0.069	29.70369
59	0	1	0	1	0.872	0.279	42.84914
60	0	0	0	1	0.220	0.081	34.72200
61	0	0	0	1	0.941	0.220	63.08514
62	0	0	0	1	0.941	0.220	55.10370
63	0	0	0	1	0.267	0.976	37.88719
64	0	1	0	1	0.267	0.116	45.99583
65	0	0	1	0	0.546	0.651	76.67378
66	1	0	1	0	0.604	0.651	59.52925
67	0	0	0	1	0.069	0.116	59.86400
68	0	0	0	1	0.069	0.116	52.82595
69	0	0	0	1	0.151	0.220	67.03763

70	0	1	0	1	0.569	0.058	34.57899
71	0	0	0	1	0.965	0.767	33.43755
72	0	0	0	0	0.418	0.325	32.23100
73	0	0	0	0	0.162	0.116	27.30386
74	0	0	0	1	0.604	0.755	54.86111
75	0	1	0	0	0.604	0.325	48.98265
76	0	0	0	0	0.313	0.511	43.06890
77	0	0	0	1	0.174	0.011	45.73675
78	0	0	1	0	0.465	0.325	49.65350
79	0	0	0	1	0.313	0.255	47.17323
80	1	0	1	0	0.546	0.906	81.64291
81	0	0	0	0	0.220	0.116	45.46628
82	0	1	0	0	0.093	0.116	39.01190
83	0	0	0	1	0.313	0.313	44.37552
84	0	1	0	0	0.186	0.267	41.90431
85	1	0	0	1	0.872	0.848	49.52411

```
candy = read.csv("candy-data.csv", row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanut	almond	nougat	crisped	rice	waffer
100 Grand	1	0	1		0	0			1
3 Musketeers	1	0	0		0	1			0
One dime	0	0	0		0	0			0
One quarter	0	0	0		0	0			0
Air Heads	0	1	0		0	0			0
Almond Joy	1	0	0		1	0			0

	hard	bar	pluribus	sugar	percent	price	percent	win	percent
100 Grand	0	1	0		0.732		0.860	66.97	173
3 Musketeers	0	1	0		0.604		0.511	67.60	294
One dime	0	0	0		0.011		0.116	32.26	109
One quarter	0	0	0		0.011		0.511	46.11	650
Air Heads	0	0	0		0.906		0.511	52.34	146
Almond Joy	0	1	0		0.465		0.767	50.34	755

what is in the dataset

Q1. How many different candy types are in this dataset?

there are 85 rows in theis dataaset

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity==1)
```

```
[1] 38
```

there are 38 fruity candy types

2.2 what is ur fav candy ?

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
      winpercent
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
candy |>
  filter(row.names(candy)=="Reese's pieces") |>
  select(winpercent)
```

```
      winpercent
Reese's pieces  73.43499
```

Reese's pieces: 73.43499 percent

Q4. What is the winpercent value for "Kit Kat"?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

the winpercent value of "tootsie roll snack bar" is 49.653503 percent

skim function

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

complete rate column

Q7. What do you think a zero and one represent for the candy\$chocolate column?

```
skim(candy$chocolate)
```

Table 3: Data summary

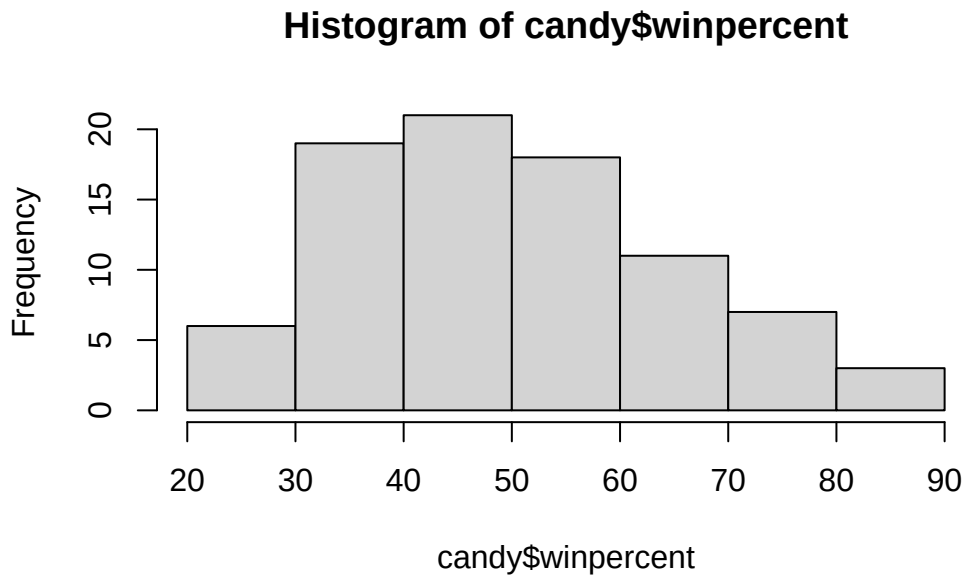
Name	candy\$chocolate
Number of rows	85
Number of columns	1
Column type frequency:	
numeric	1
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
data	0	1	0.44	0.5	0	0	0	1	1	

exploratory analysis

```
hist(candy$winpercent)
```

Q9. Is the distribution of winpercent values symmetrical?

no, it is skewed to right

Q10. Is the center of the distribution above or below 50%?

above 50% it is 50.3167638

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

steps to solve this: 1. find all choco candy in the dataset 2. extract or find their winpercent values 3. calculate the mean of these values 4. do the same for fruit

```
chocolate <- mean(candy$winpercent[as.logical(candy$chocolate)])
fruity<- mean(candy$winpercent[as.logical(candy$fruity)])
chocolate
```

```
[1] 60.92153
```

```
fruity
```

```
[1] 44.11974
```

on average, chocolate has a higher ranking than fruity candy.

Q12. Is this difference statistically significant?

```
chocolate.win <- candy$winpercent[as.logical(candy$chocolate)]
fruity.win <- candy$winpercent[as.logical(candy$fruity)]
t.test(chocolate.win, fruity.win)
```

Welch Two Sample t-test

```
data: chocolate.win and fruity.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

five least disliked candy types are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

```
inds <- order(candy$winpercent)
head(candy[inds,],n=5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0

	crispedricewafer	hard	bar	pluribus	sugarpercent	pricepercent
Nik L Nip	0	0	0	1	0.197	0.976
Boston Baked Beans	0	0	0	1	0.313	0.511
Chiclets	0	0	0	1	0.046	0.325
Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q14. What are the top 5 all time favorite candy types out of this set?

```
inds <- order(candy$winpercent)
tail(candy[inds,],n=5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Snickers	1	0	1	1	1
Kit Kat	1	0	0	0	0
Twix	1	0	1	0	0
Reese's Miniatures	1	0	0	1	0
Reese's Peanut Butter cup	1	0	0	1	0

	crispedricewafer	hard	bar	pluribus	sugarpercent
Snickers	0	0	1	0	0.546
Kit Kat	1	0	1	0	0.313
Twix	1	0	1	0	0.546
Reese's Miniatures	0	0	0	0	0.034
Reese's Peanut Butter cup	0	0	0	0	0.720

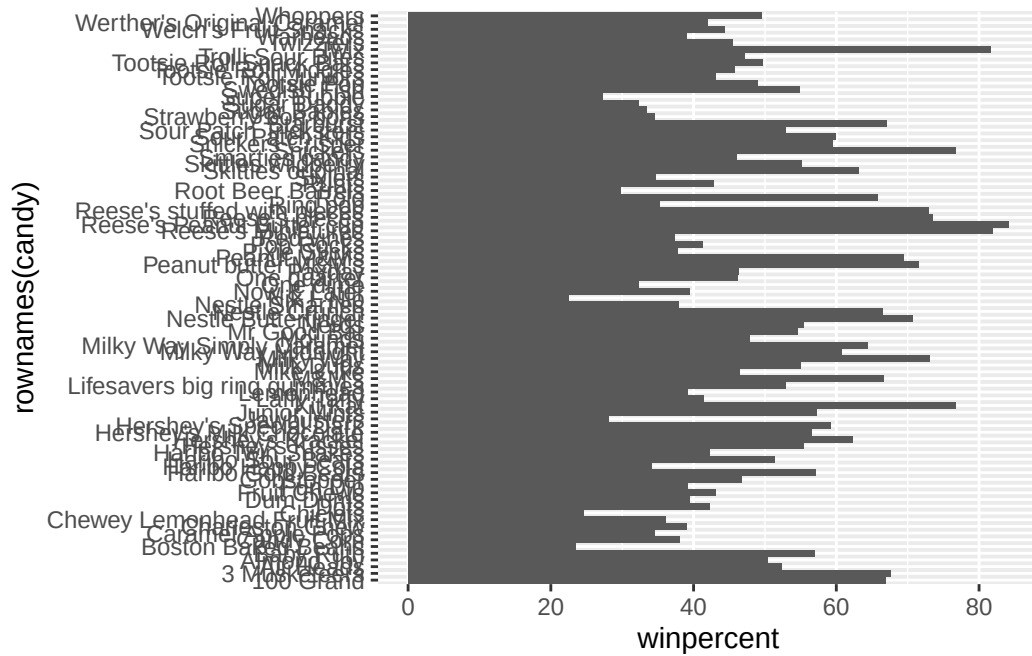
	pricepercent	winpercent
Snickers	0.651	76.67378
Kit Kat	0.511	76.76860
Twix	0.906	81.64291
Reese's Miniatures	0.279	81.86626
Reese's Peanut Butter cup	0.651	84.18029

Top five are Snickers, Kit Kat, Twix, Reese's Miniatures, Reese's Peanut Butter cup

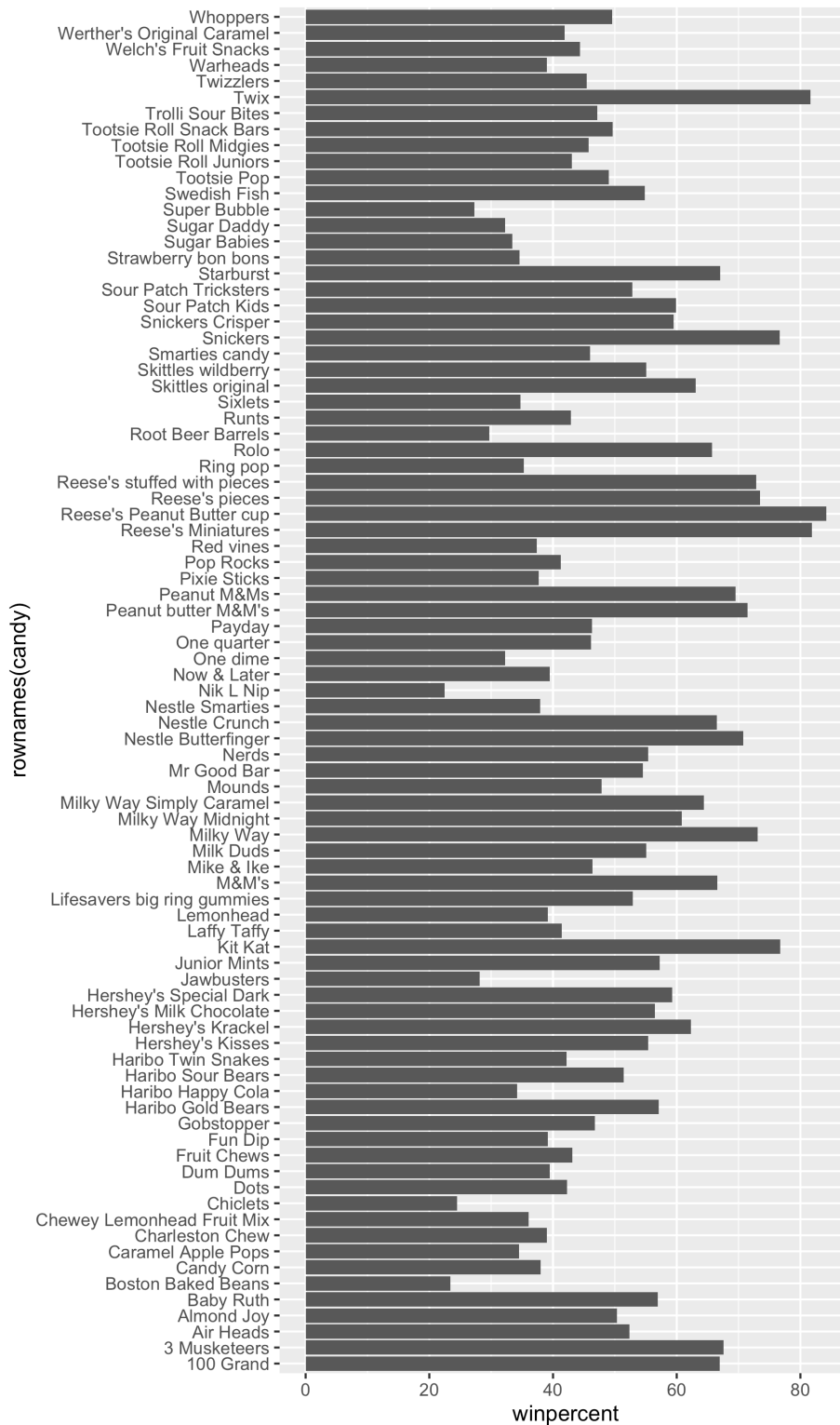
Q15. Make a first barplot of candy ranking based on winpercent values.

```
library(ggplot2)

ggplot(candy)+
  aes(winpercent, rownames(candy))+
  geom_col()
```



```
ggsave("barplot1.png", height=10, width=6)
```

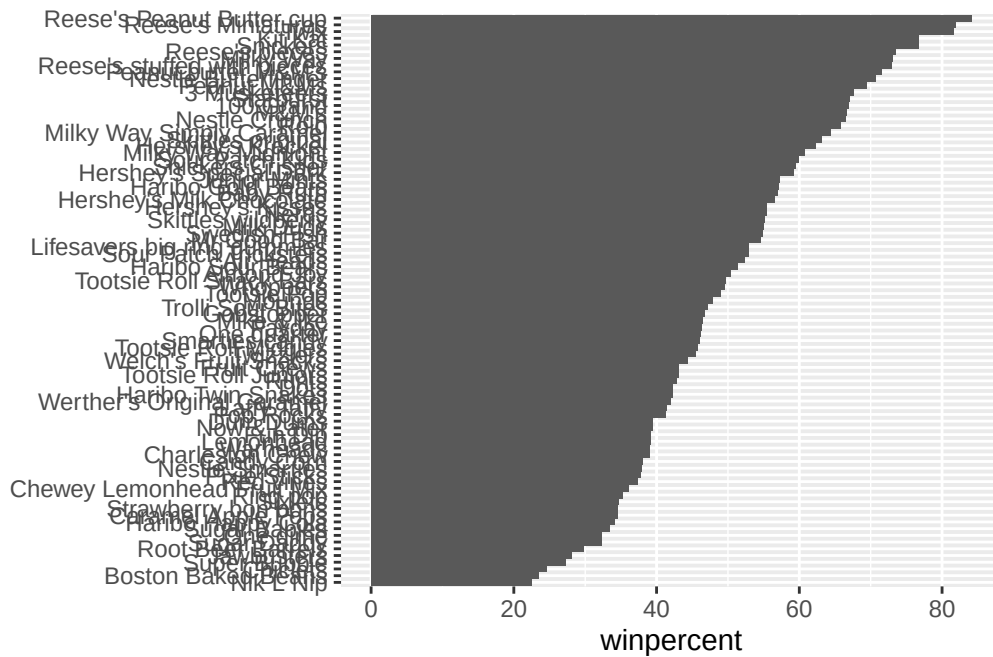


>Q16. This is quite

ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
library(ggplot2)

ggplot(candy)+
  aes(winpercent,
      reorder(rownames(candy), winpercent))+
  geom_col()+
  ylab("") #turn off y label we don't need that
```



```
ggsave("barplot2.png", height=10, width=6)
```

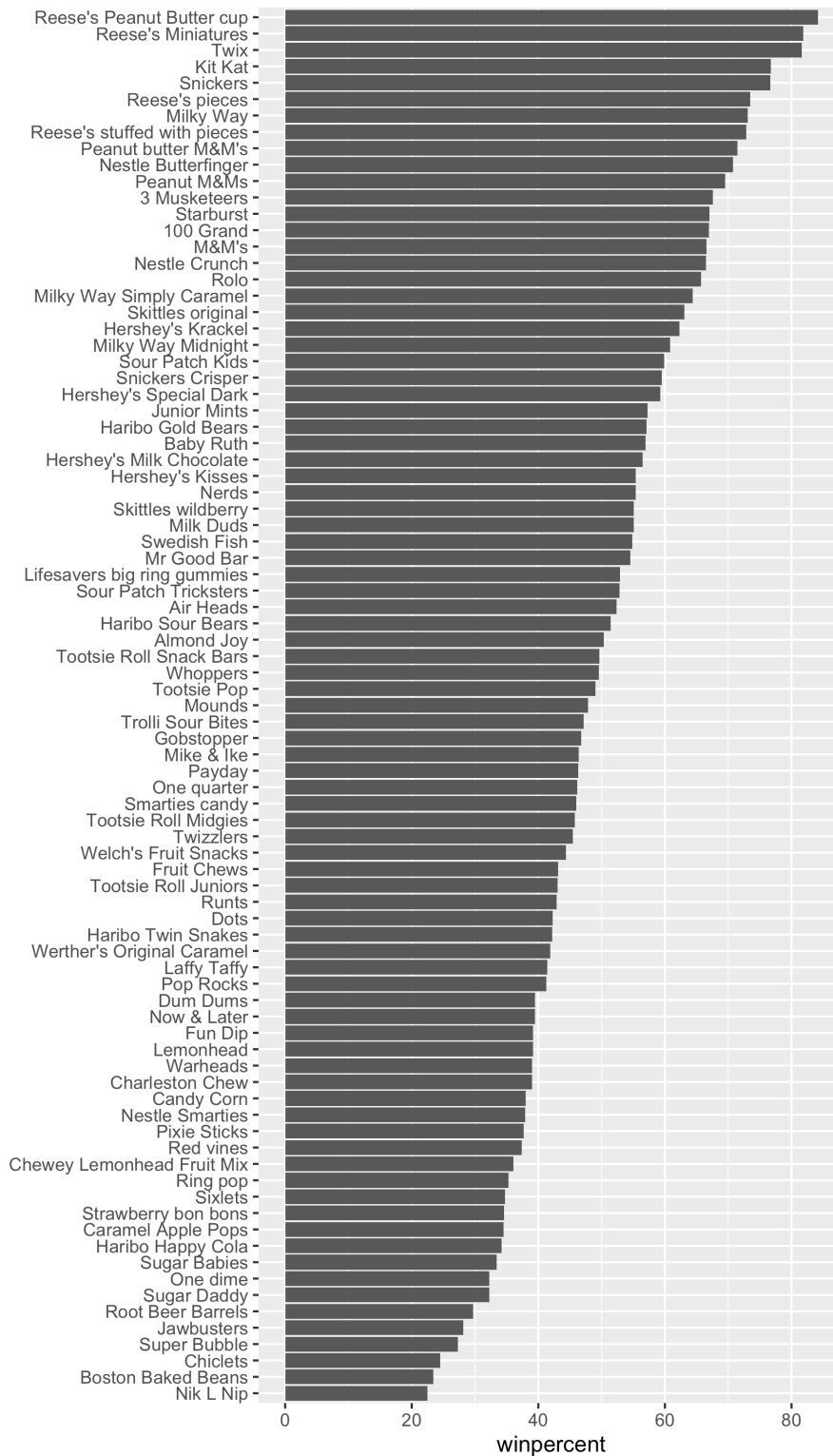


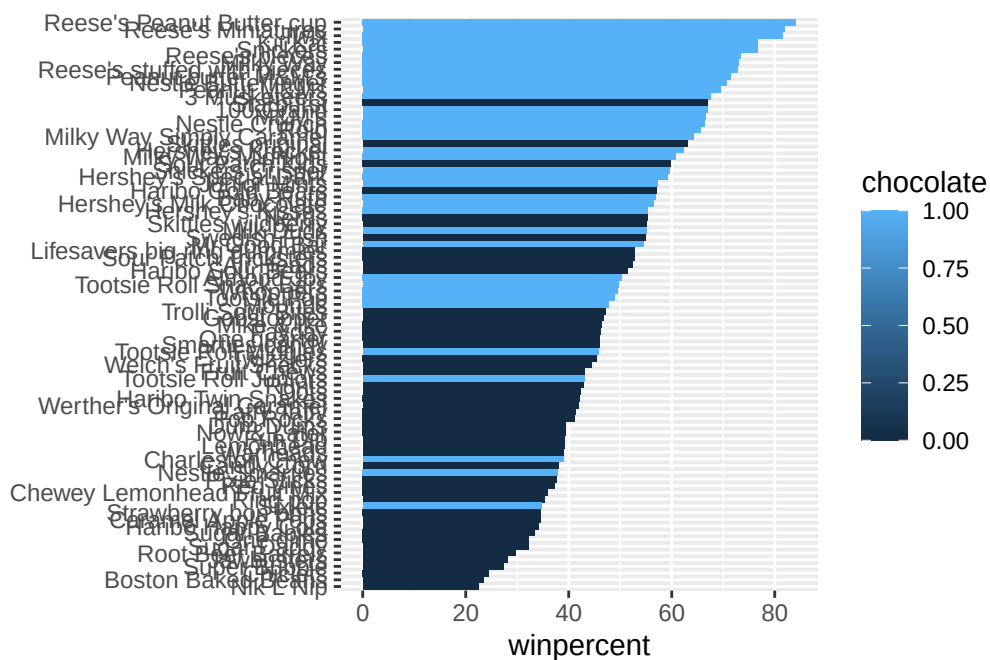
Figure 1: My 2nd barplot for Q16

time to add some useful color

color by chocolate:

```
library(ggplot2)

ggplot(candy)+
  aes(winpercent,
      reorder(rownames(candy), winpercent),
      fill=chocolate)+
  geom_col()+
  ylab("")
```



I want custom colors that i pick so we need to make this ourselves...

```
my_cols <- rep("black", nrow(candy))
my_cols[candy$chocolate==1]<-"chocolate"
my_cols[candy$bar==1]<-"blue"
my_cols[candy$fruity==1] <-"hotpink"
my_cols
```

```
[1] "blue"      "blue"      "black"     "black"     "hotpink"   "blue"
```



```
ggplot(candy)+
  aes(winpercent,
      reorder(rownames(candy), winpercent),
      fill=chocolate)+
  geom_col(fill=my_cols)+
  ylab("")
```



```
ggsave("barplot3.png", height=10, width=6)
```

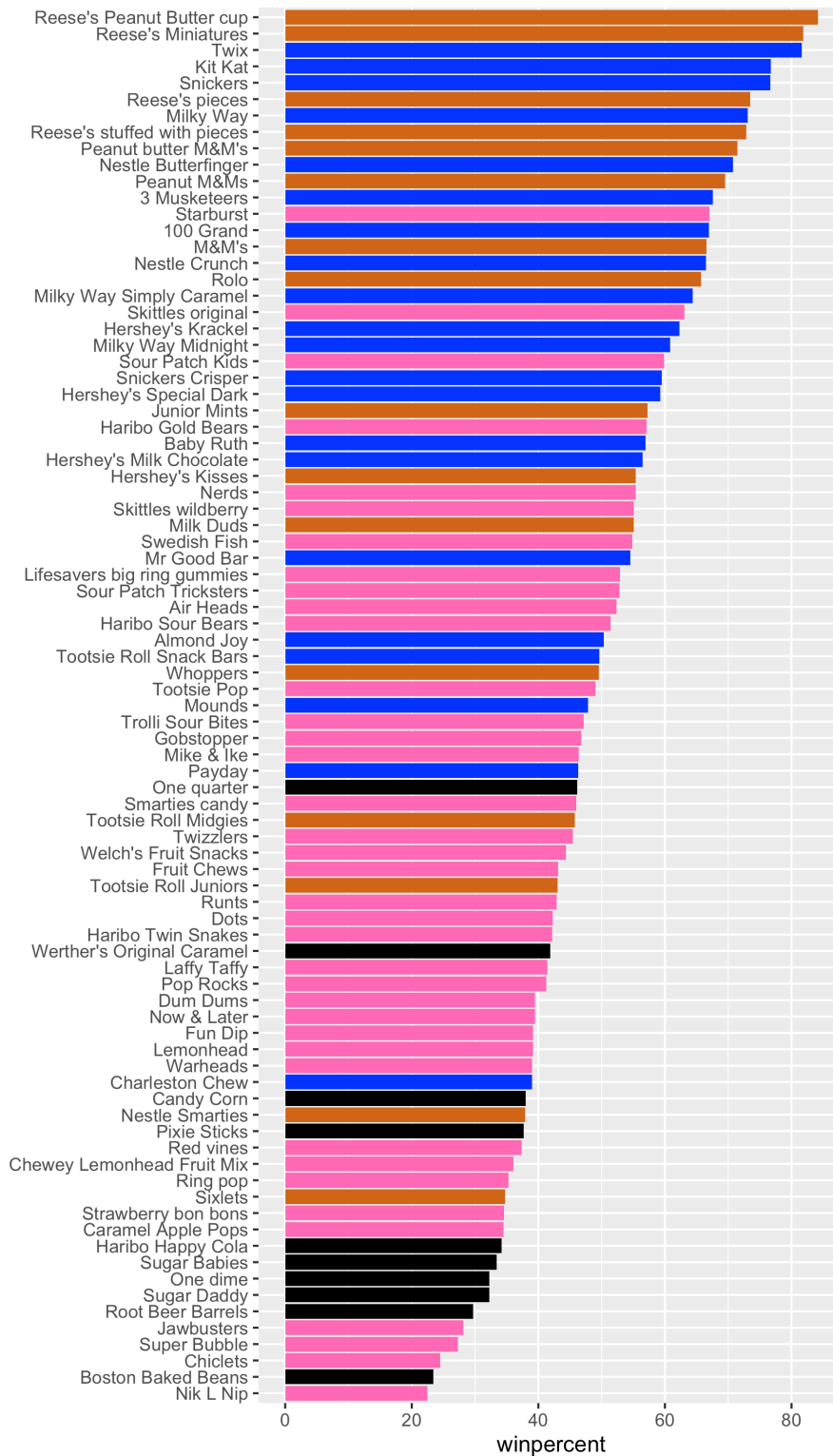


Figure 2: My barplot colored

Q17. What is the worst ranked chocolate candy?

it is sixlets

Q18. What is the best ranked fruity candy?

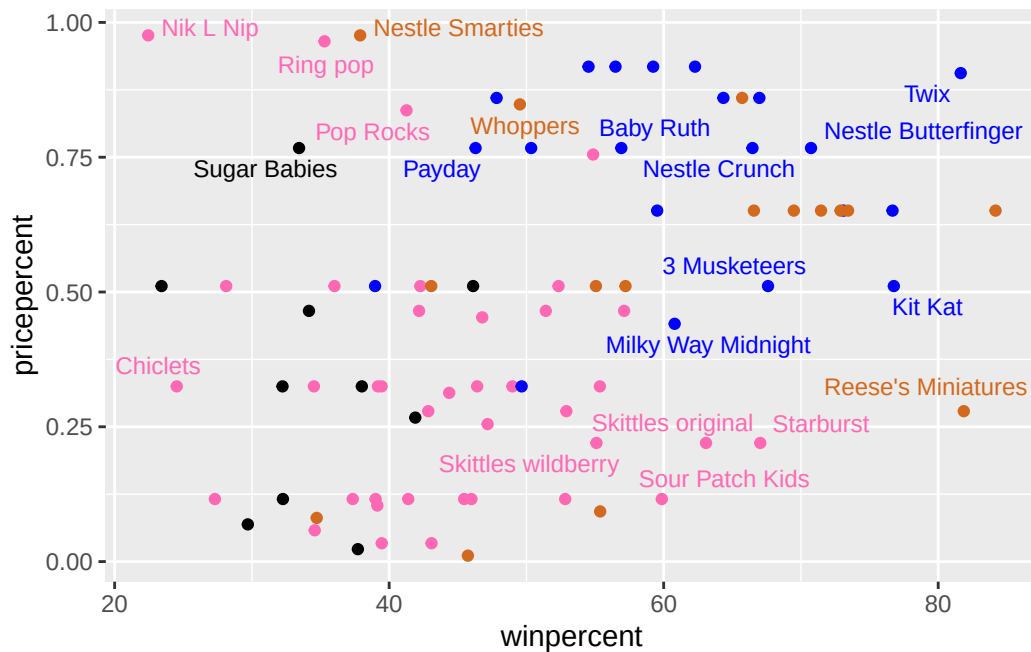
it is the pink one so it is starbursts

Taking a look at pricepercent

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(x=winpercent, y=pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```

Warning: ggrepel: 65 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

reese's miniatures

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

most expensive are hershey's milk chocolate, hershey's krackel, ring pop, nestle smarties, and nik l nip

least popular is nik l nip

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

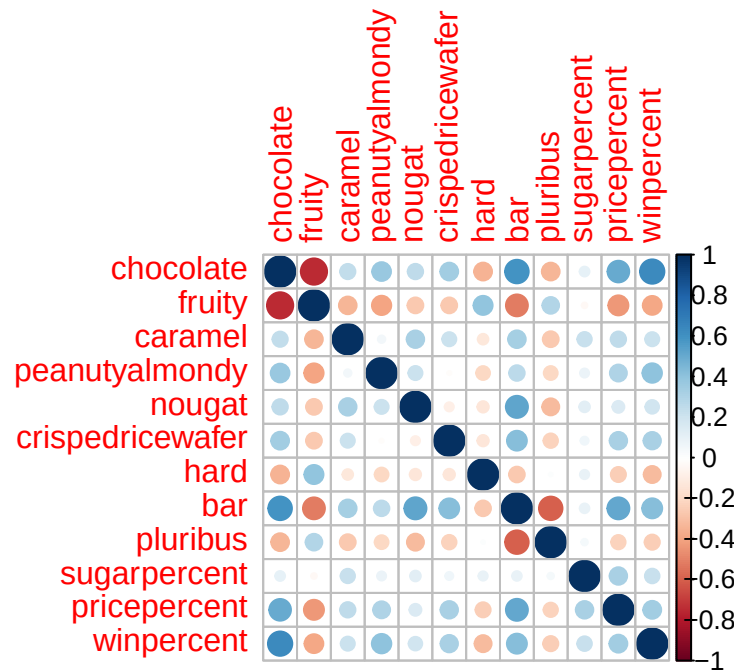
exploring the correlation structure

pearson correlation values range from -1 to +1

```
library(corrplot)
```

corrplot 0.95 loaded

```
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

anti-correlated is fruity vs chocolate

Q23. Similarly, what two variables are most positively correlated?

most positively correlated are variables that are the same as each other like chocolate and chocolate

PCA

REMEMBER TO USE SCALE=TRUE

```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

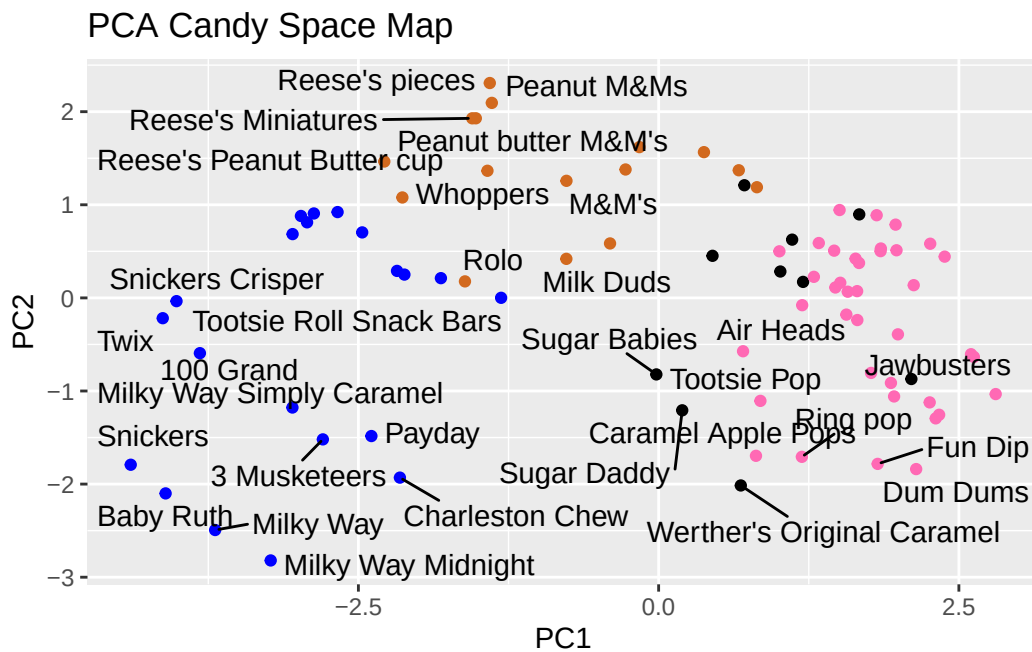
	PC8	PC9	PC10	PC11	PC12
--	-----	-----	------	------	------

Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

the main result figure: the PCA score plot:

```
ggplot(pca$x)+
  aes(PC1, PC2, label=rownames(pca$x))+
  geom_point(col=my_cols)+
  geom_text_repel()+
  labs(title="PCA Candy Space Map")
```

Warning: ggrepel: 54 unlabeled data points (too many overlaps). Consider increasing max.overlaps

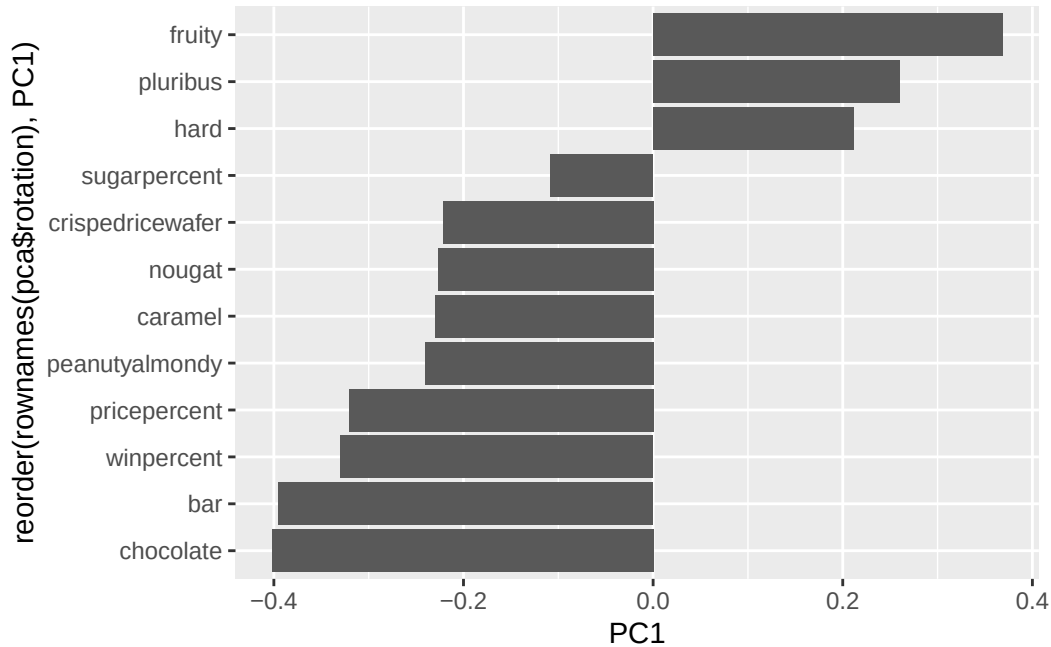


Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

fruity, pluribus, and hard

The "loadings" plot for pC1

```
ggplot(pca$rotation)+
  aes(PC1,reorder(rownames(pca$rotation), PC1))+
  geom_col()
```



Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

it would be chocolate, bar arrangement, cheap. PCA showed correlation between price points and preferences. The bar graphs showed which ones are most popular. correlation showed the different combination that people like and dislike.